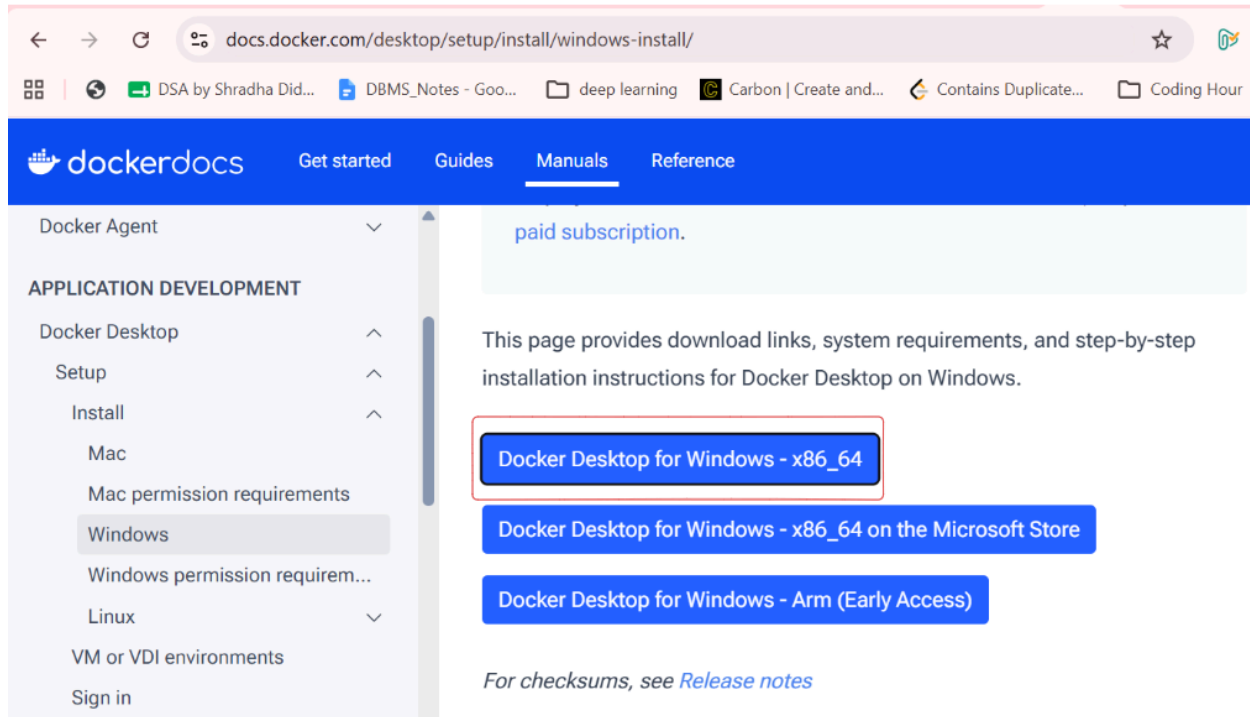


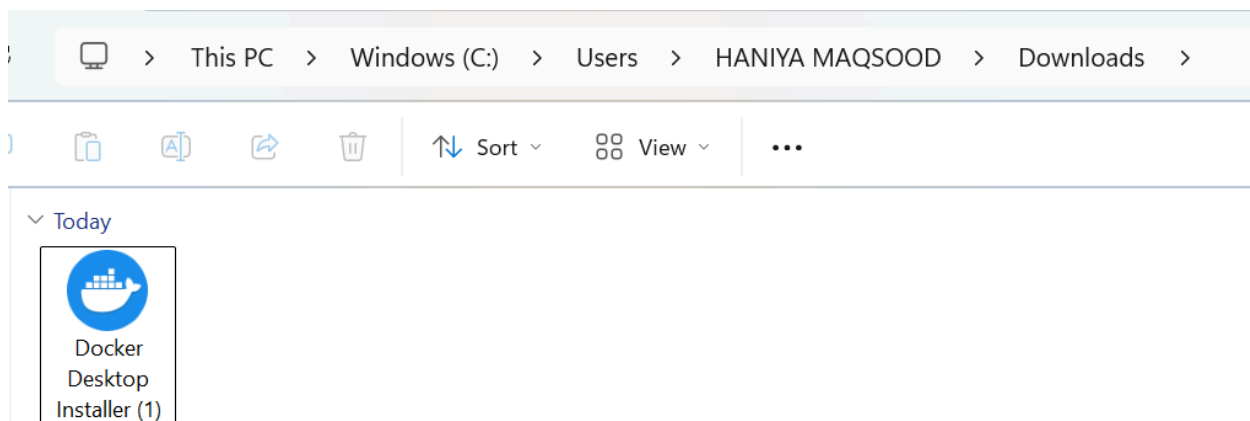
DOCKER DESKTOP INSTALLATION

Go to this link:

<https://docs.docker.com/desktop/setup/install/windows-install/>



After downloading the setup file. Open Download folder and double click on **Docker Desktop exe** file



Click next and install the **Docker Desktop** using default configuration and restart your **PC**.

Open **Powershell** as an **Administrator** and **type docker version**. It should shows below output

```
Administrator: Windows PowerShell

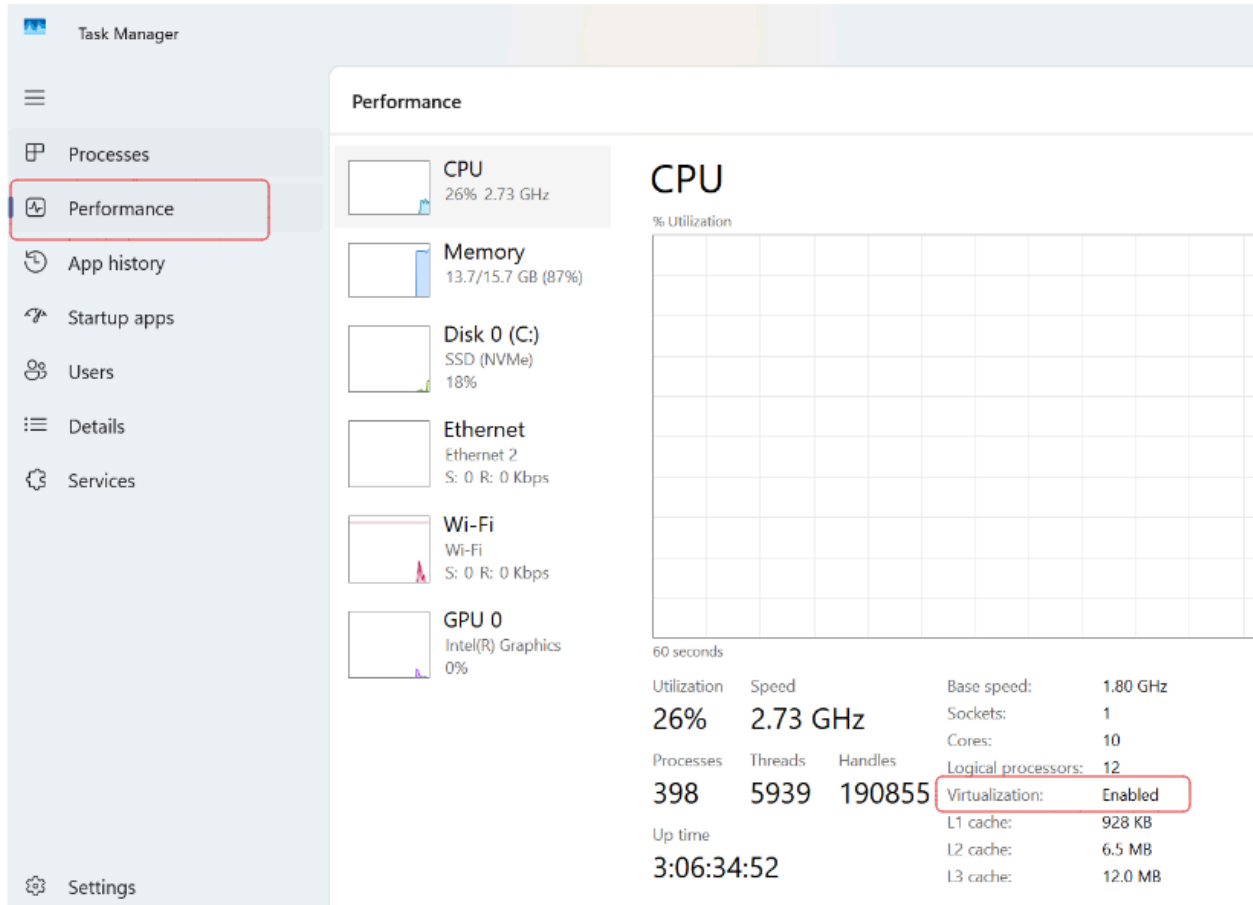
Windows version: 10.0.26200.8875
PS C:\WINDOWS\system32> docker version
Client:
Version:           28.0.1
API version:       1.48
Go version:        go1.23.6
Git commit:        068a01e
Built:             Wed Feb 26 10:41:52 2025
OS/Arch:           windows/amd64
Context:           desktop-linux

Server: Docker Desktop 4.39.0 (184744)
Engine:
Version:           28.0.1
API version:       1.48 (minimum version 1.24)
Go version:        go1.23.6
Git commit:        bbd0a17
Built:             Wed Feb 26 10:41:16 2025
OS/Arch:           linux/amd64
Experimental:      false
containerd:
Version:           1.7.25
GitCommit:         bcc810d6b9066471b0b6fa75f557a15a1cbf31bb
runc:
Version:           1.2.4
GitCommit:         v1.2.4-0-g6c52b3f
docker-init:
Version:           0.19.0
GitCommit:         de40ad0
```

Now Docker is installed.

CHECK VIRTUALIZATION IS ENABLED?

Go to the Task manager and then the performance tab and check virtualization on the right corner .



CHECK WSL IS INSTALLED?

Open Powershell as administrator mode and type [wsl --version]

```
PS C:\WINDOWS\system32> wsl --version
WSL version: 2.4.13.0
Kernel version: 5.15.167.4-1
WSLg version: 1.0.65
MSRDC version: 1.2.5716
Direct3D version: 1.611.1-81528511
DXCore version: 10.0.26100.1-240331-1435.ge-release
Windows version: 10.0.26200.8875
```

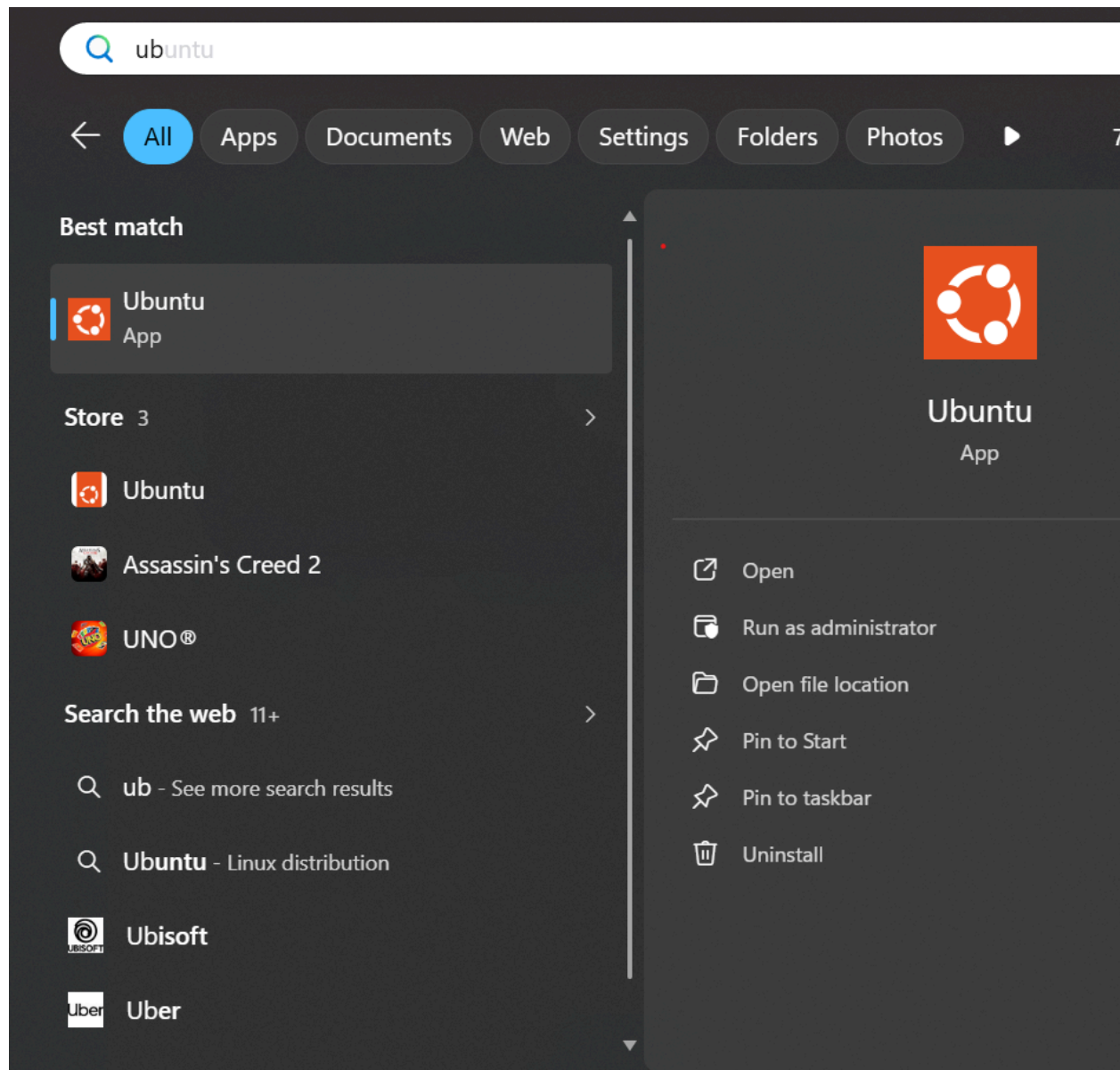
If it's not already installed. Type command

```
wsl --install -d Ubuntu
```

After installation. Check the version.

Check Ubuntu is installed?

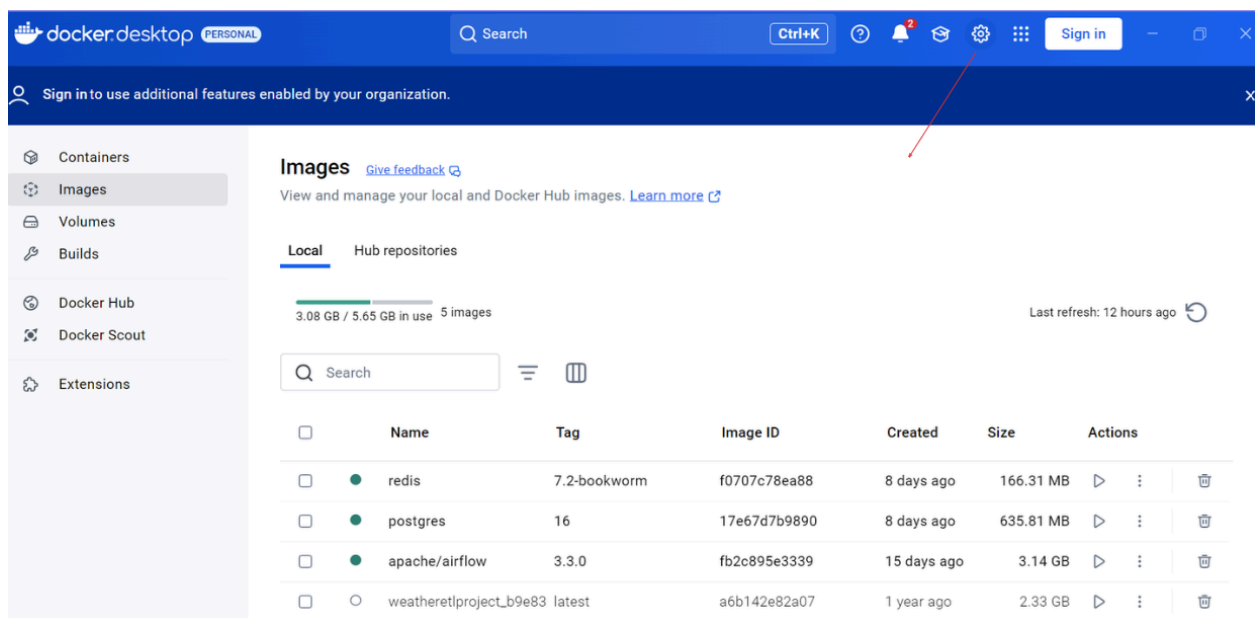
Click on the start button and search **Ubuntu** and open it.

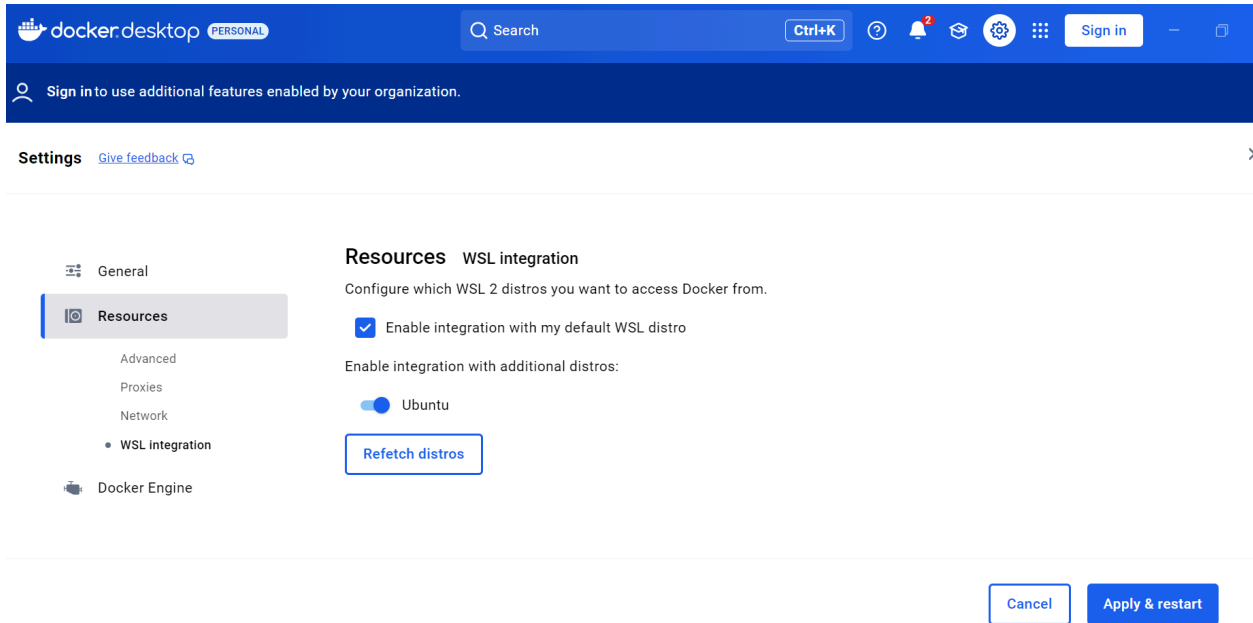


If it's the first time, you need to set the username and password. This is a Linux operating system. When you type the password. The cursor will not move, just enter the password and click enter and re-write the password.

Enable Docker in WSL

1. Open Docker Desktop → Settings → Resources → WSL Integration
2. Enable integration with Ubuntu
3. Click Apply & Restart





Inside Ubuntu type:
docker --version
docker compose version

```
haniya@DESKTOP-U0189K2:~$ docker version
Client:
Version:           28.0.1
API version:       1.48
Go version:        go1.23.6
Git commit:        068a01e
Built:             Wed Feb 26 10:40:04 2025
OS/Arch:           linux/amd64
Context:           default

Server: Docker Desktop 4.0.0 ( )
Engine:
Version:           28.0.1
API version:       1.48 (minimum version 1.24)
Go version:        go1.23.6
Git commit:        bbd0a17
Built:             Wed Feb 26 10:41:16 2025
OS/Arch:           linux/amd64
Experimental:      false
containerd:
Version:           1.7.25
GitCommit:         bcc810d6b9066471b0b6fa75f557a15a1cbf31bb
runc:
Version:           1.2.4
GitCommit:         v1.2.4-0-g6c52b3f
docker-init:
Version:           0.19.0
GitCommit:         de40ad0
haniya@DESKTOP-U0189K2:~$ |
```

The output should be like this

Create a Working Directory

Inside Ubuntu type this command and create the folder.

```
mkdir airflow-docker
cd airflow-docker
mkdir -p ./dags ./logs ./plugins ./config
```

Type **ls** to check the folder has been created.

```
haniya@DESKTOP-U0189K2:~$ ls
airflow  airflow-docker  airflow_project
haniya@DESKTOP-U0189K2:~$ cd airflow-docker
haniya@DESKTOP-U0189K2:~/airflow-docker$ ls
config  dags  docker-compose.yaml  logs  plugins
```

Download Docker Compose File

```
curl -Lfo 'https://airflow.apache.org/docs/apache-airflow/stable/docker-compose.yaml'
```

Create the .env File

```
echo -e "AIRFLOW_UID=50000" > .env
```

Initialize Airflow

```
docker compose up airflow-init
```

Wait for it to finish (you'll see `exited with code 0`), then press `Ctrl + C`.


```

haniya@DESKTOP-U0189W2:~/airflow-docker$ docker compose up airflow-init
WARN[0000] The "FERNET_KEY" variable is not set. Defaulting to a blank string.
WARN[0000] The "FERNET_KEY" variable is not set. Defaulting to a blank string.
WARN[0000] The "FERNET_KEY" variable is not set. Defaulting to a blank string.
WARN[0000] The "FERNET_KEY" variable is not set. Defaulting to a blank string.
WARN[0000] The "FERNET_KEY" variable is not set. Defaulting to a blank string.
WARN[0000] The "FERNET_KEY" variable is not set. Defaulting to a blank string.
WARN[0000] The "FERNET_KEY" variable is not set. Defaulting to a blank string.
WARN[0000] The "FERNET_KEY" variable is not set. Defaulting to a blank string.
[+] Running 3/3
✔ Container airflow-docker-postgres-1      Running      0.0s
✔ Container airflow-docker-redis-1         Running      0.0s
✔ Container airflow-docker-airflow-init-1   Created      0.0s
Attaching to airflow-init-1
airflow-init-1 |
airflow-init-1 | Creating missing opt dirs if missing:
airflow-init-1 |
airflow-init-1 |
airflow-init-1 | Airflow version:
airflow-init-1 | The container is run as root user. For security, consider using a regular user account.
airflow-init-1 | 2026-07-21T20:38:44.462935Z [info] Performing upgrade to the metadata database [airflow.cli.commands.db_comman
d] loc=db_command.py:134 url=postgresql+psycopg2://airflow:***@postgres/airflow
airflow-init-1 | 2026-07-21T20:38:44.632018Z [info] ] setup plugin alembic.autogenerate.schemas [alembic.runtime.plugins] loc=plu
gins.py:37
airflow-init-1 | 2026-07-21T20:38:44.632273Z [info] ] setup plugin alembic.autogenerate.tables [alembic.runtime.plugins] loc=plug
ins.py:37
airflow-init-1 | 2026-07-21T20:38:44.633881Z [info] ] setup plugin alembic.autogenerate.types [alembic.runtime.plugins] loc=plugi
ns.py:37
airflow-init-1 | 2026-07-21T20:38:44.635585Z [info] ] setup plugin alembic.autogenerate.constraints [alembic.runtime.plugins] loc
=plugins.py:37
airflow-init-1 | 2026-07-21T20:38:44.635830Z [info] ] setup plugin alembic.autogenerate.defaults [alembic.runtime.plugins] loc=pl

```

For the first time. Output will only be Running, Created and Started.

Start Airflow

`docker compose up -d`

Open the UI

Go to <http://localhost:8080>

Username: airflow

password : airflow

Notes: There might be permission denied issues when you run any command. So for permission issues you can run the following command.

```
sudo chmod -R 777 ./config
```

While you are doing everything on Ubuntu. Just make sure Docker Desktop should be in running state